the electrodes. Each electrode set has the same area of A_e, $A = \varepsilon_0 A_e$, where ε_0 is the permeability of free space.

In the general case, the mutual inductance M is a complex non-analytical function. This represents the main difficulty for the analytical study of the suspension model (2). However, we can account for some particularities of the micro-machined device, in that the linear sizes of the coils and proof mass are much larger than the levitation height, h_l, and the distribution of the density of the induced eddy current is not homogenous. The induced eddy current is distributed along the levitated proof mass in such a way that two circuits corresponding to maximum values of the eddy current density can be identified, as shown in Figure 4a. The eddy current circuit i_{es} is defined geometrically as a circle having the same diameter as the proof mass. The second circuit i_{el} is also a circle, but with the diameter of the levitation coil [36]. Actually, the second circuit can be considered as the current image of the levitation coil. Due to the mentioned particularities of the device, the force interaction along the vertical direction is reduced to an interaction between eddy current i_{el} and the levitation coil current [40]. Considering both the levitation coil and the eddy current circuit as filamentary circles, the mutual inductance between the levitation coil and eddy current can be described by the Maxwell formula ([41], page 6); thus:

$$k^2 = \frac{4r_l^2}{4r_l^2 + (h_l + y)^2}; \quad M = \mu_0 r_l \left[\left(\frac{2}{k} - k \right) K(k) - \frac{2}{k} E(k) \right], \tag{3}$$

where μ_0 is the magnetic permeability of free space, r_l is the radius of the levitation coil and K and E are complete elliptic integrals of the first and second kinds [42]. Then, accounting for (3), Model (2) becomes:

$$\begin{aligned} &m\frac{d^2 q_v}{dt^2} + mg - \frac{I^2 a^2}{L} \left[\left(\frac{2}{k} - k \right) K(k) - \frac{2}{k} E(k) \right] \frac{2}{k^2} \\ &\times \left[\frac{2 - k^2}{2(1 - k^2)} E(k) - K(k) \right] \cdot \frac{\xi^2 (1 + \frac{q_v}{h_l})}{h_l (1 + \xi^2 (1 + \frac{q_v}{h_l})^2)^{3/2}} - \frac{AU^2 q_v}{(h - q_v^2)^2} = F_v, \end{aligned} \tag{4}$$

where $a = r_l \mu_0$ and $\xi = h_l / (2r_l)$. Model (4) is analytical, nonlinear and quasi-exact, but due to the elliptic integrals, it can be studied only numerically. For further analysis, Model (4) is presented in dimensionless form as follows:

$$\begin{aligned} &\frac{d^2 \lambda}{d\tau^2} + 1 - \eta \left[\left(\frac{2}{k} - k \right) K(k) - \frac{2}{k} E(k) \right] \frac{2}{k^2} \\ &\times \left[\frac{2 - k^2}{2(1 - k^2)} E(k) - K(k) \right] \cdot \frac{\xi^2 (1 + \lambda) \kappa}{(1 + \xi^2 (1 + \lambda)^2)^{3/2}} - \frac{\beta \lambda}{(1 - \lambda^2)^2} = \tilde{F}, \end{aligned} \tag{5}$$

where $\tau = \sqrt{g/h}t$, $\lambda = q_v / h$, $\eta = I^2 a^2 / (mghL)$, $\beta = AU^2 / (mgh^2)$, $\kappa = h / h_l$ and $\tilde{F} = F_v / mg$.

Moreover, upon ensuring a condition described further below, Equation (3) can be approximated well by the logarithmic function [34]:

$$M = \mu_0 r_l \left[\ln \frac{4}{\xi (1 + y/h_l)} - 2 \right]. \tag{6}$$

Hence, accounting for the latter equation, the following reduced analytical model of a suspension is proposed:

$$m\frac{d^2 y}{dt^2} + mg - \frac{I^2 a^2}{L} \frac{1}{h_l + y} \left[\ln \frac{4}{\xi (1 + y/h_l)} - 2 \right] - \frac{AU^2 q_v}{(h^2 - q_v^2)^2} = F_v. \tag{7}$$

In dimensionless form, Equation (7) becomes:

$$\frac{d^2\lambda}{d\tau^2} + 1 - \eta\frac{\kappa}{1+\kappa\lambda}\left[\ln\frac{4}{\xi(1+\kappa\lambda)} - 2\right] - \frac{\beta\lambda}{(1-\lambda^2)^2} = \tilde{F}. \tag{8}$$

As is shown in Appendix A below, the accuracy of approximation of modeling the electromagnetic force is dependent on the parameter ξ. If ξ is less than 0.3, the electromagnetic force is approximated by the logarithmic function (6) with an error less than 6%. When parameter ξ vanishes, the error between the exact Equation (3) and the approximation (6) also vanishes. It is worth noting that, for all known prototypes of μ-HCS published in the literature, parameter ξ is less than 0.25. This fact indicates the applicability of the reduced model for further analytical study of μ-HCS, as has already been successfully demonstrated for instance in [32]. Hence, Model (8) is the main framework for further analysis of the static and dynamic pull-in.

4. Static Pull-In Instability

We now study the load-free behavior of the device upon changing the strength of the electric field, characterized by the dimensionless parameter β (dimensionless squared voltage). For this reason, Equation (8) is written as a set in terms of the phase coordinates [31]:

$$\begin{cases} \dfrac{d\lambda}{d\tau} = \omega; \\ \dfrac{d\omega}{d\tau} = -1 + \eta\dfrac{\kappa}{1+\kappa\lambda}\left[\ln\dfrac{4}{\xi(1+\kappa\lambda)} - 2\right] + \dfrac{\beta\lambda}{(1-\lambda^2)^2}. \end{cases} \tag{9}$$

From (9), the equilibrium state of the system can be defined as:

$$f(\lambda,\beta) = -1 + \eta\frac{\kappa}{1+\kappa\lambda}\left[D - \ln(1+\kappa\lambda)\right] + \frac{\beta\lambda}{(1-\lambda^2)^2}, \tag{10}$$

where $D = \ln\frac{4}{\xi} - 2$ is the design parameter depending on ξ. At the equilibrium point $\lambda = 0$, the function f must equal zero; this point requires that parameter $\eta = 1/D$. Hence, the static equilibrium state of the system, which relates the vertical coordinate with the strength of the electric field, is:

$$f(\lambda,\beta) = -\frac{\kappa\lambda}{1+\kappa\lambda} - \frac{\ln(1+\kappa\lambda)}{D(1+\kappa\lambda)} + \frac{\beta\lambda}{(1-\lambda^2)^2} \equiv 0. \tag{11}$$

Since the vertical displacement of the proof mass is limited by the positions of the top and bottom electrodes, the variable λ is varied within a range of $-1 \le \lambda \le 1$. Furthermore, taking into account that constant D and κ can be considered within the following ranges of $1 < D < 4.0$ and $0 < \kappa \le 1$, the bifurcation diagram, which relates the distribution of saddles (unstable equilibrium), centers (stable equilibrium) and bifurcations with the dimensionless square voltage β is shown in Figure 5a. Note that the presented prototype of the hybrid suspension in Section 2 has the following values of dimensionless parameters, namely $D = 2.0456$, $\xi = 0.07$ and $\kappa = 0.3333$. Analysis of the diagram shows that two bifurcation points can be recognized, denoted as A and B. Both bifurcation points correspond to the pull-in instability. This means that, once the strength of the electric field has achieved the value characterized by β_A, the proof mass is pulled in and moves toward the top electrodes. At point B, where the strength of the electric field is characterized by β_B, the proof mass at the position characterized by λ_B is also pulled in, but moves already toward the bottom electrodes.

The bifurcation point A is defined by the following parameters:

$$\lambda_A = 0; \;\; \beta_A = \kappa(1 + 1/D). \tag{12}$$

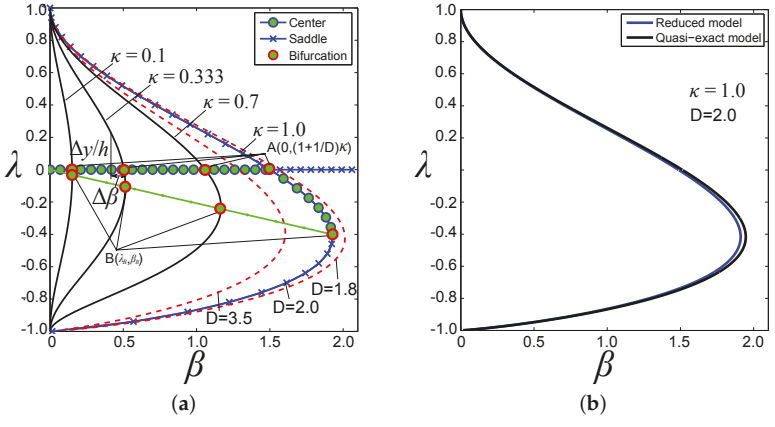


Figure 5. Bifurcation diagram: (**a**) dashed red lines show the evolution of the bifurcation map depending on constant D ($\kappa = 1.0$); solid lines depict the evolution of the bifurcation map depending on spacing $\kappa = h/h_l$ ($D = 2.0$); (**b**) comparison of the quasi-exact and reduced models for $D = 2.0$, $\kappa = 1.0$ and $\xi = 0.07$ (the relative error is less than 2%).

Parameters β_B and λ_B characterizing bifurcation point B (static pull-in instability) are defined numerically as the solution of the following set of equations:

$$\begin{cases} -3D\kappa^2\lambda^4 - \kappa(4D+1)\lambda^3 - \kappa^2 D\lambda^2 + \kappa\lambda - (2\kappa\lambda^3 + 3\lambda^2 + 2\kappa\lambda + 1)\ln(1+\kappa\lambda) = 0; \\ \beta = \left(\kappa\lambda + \dfrac{\ln(1+\kappa\lambda)}{D}\right)\dfrac{(1-\lambda^2)}{\lambda(1+\kappa\lambda)}. \end{cases} \tag{13}$$

when κ is small, then the set (13) has an approximate solution:

$$\lambda_B \approx -\frac{\kappa}{4}\frac{D+3/2}{D+1};\ \ \beta_B \approx \kappa(1+1/D)\left(1+\frac{\kappa^2}{4}\frac{D+3/2}{D+1}\right). \tag{14}$$

In addition, a comparison between reduced Model (8) and the quasi-exact model (5) is performed for the considered design of the hybrid suspension in this work, characterized by the following dimensionless parameters $D = 2$ and $\xi = 0.07$. The result of this comparison is presented in Figure 5b. Analysis of Figure 5b reveals that the relative error is not in excess of 2%.

The ranges of parameters $\lambda = 0$, $0 \le \beta < \beta_A$ and $-\lambda_B < \lambda < 0$ and $\beta_A \le \beta < \beta_B$ establish a stable state of equilibrium (see Figure 5a). A region near the bifurcation point A is of special interest, because it defines a state of zero stiffness of the suspension. As seen, a decrease of stiffness leads to decreasing a range of linear displacement of the proof mass. Near bifurcation point A, the range of displacement becomes:

$$h\frac{\Delta\beta}{\kappa(1+1/D)} = h\frac{\kappa(1+1/D)-\beta}{\kappa(1+1/D)} \ge \triangle y. \tag{15}$$

Using Equation (15), the minimum possible value of linear stiffness still capable of upholding stable levitation can be estimated. For instance, in the fabricated design of the suspension (see Table 1), upon controlling the linear displacement of the proof mass Δy within a range of ± 1 µm, the relative minimization of the stiffness can be expected to be around 0.007. This means that the initial stiffness generated by the inductive suspension can be reduced by two orders of magnitude. Note that the design of the suspension corresponds to a bifurcation curve with $\kappa = 0.3333$, as shown in Figure 5a.

5. Dynamic Pull-In Instability

An equation for the integral curves of set (9) can be obtained as follows:

$$\frac{d\omega}{d\lambda} = \frac{-\dfrac{\kappa\lambda}{1+\kappa\lambda} - \dfrac{\ln(1+\kappa\lambda)}{D(1+\kappa\lambda)} + \dfrac{\beta\lambda}{(1-\lambda^2)^2}}{\omega}. \tag{16}$$

Integrating (16), the equation of energy is obtained as:

$$\omega^2 + 2\lambda - 2\frac{\ln(1+\kappa\lambda)}{\kappa} + \frac{\ln^2(1+\kappa\lambda)}{D\kappa} - \frac{\beta}{1-\lambda^2} = G, \tag{17}$$

where G is an arbitrary constant of the integration. From the analysis of (17), it is very important to note the following observation, that in order to operate the device properly, it is required to remove the energy of the electric field characterized by parameter β from the system, in order to satisfy the initial conditions. Since G is an arbitrary constant, it can be chosen to be equal to $-\beta$. Then, the final form of the integral equation becomes:

$$\omega^2 + 2\lambda - 2\frac{\ln(1+\kappa\lambda)}{\kappa} + \frac{\ln^2(1+\kappa\lambda)}{D\kappa} - \frac{\beta\lambda^2}{1-\lambda^2} = G'. \tag{18}$$

From (18), the dynamic equilibrium state can be written as:

$$f_d(\lambda,\beta) = 2\lambda - 2\frac{\ln(1+\kappa\lambda)}{\kappa} + \frac{\ln^2(1+\kappa\lambda)}{D\kappa} - \frac{\beta\lambda^2}{1-\lambda^2} \equiv 0. \tag{19}$$

Using (19), the bifurcation diagram can be plotted as shown in Figure 6. Similar to the static bifurcation diagram, it has two pull-in instability points (see Figure 6). One point has the same coordinates as point A shown in (12) corresponding to the static pull-in instability, but B_d has different coordinates compared to the static pull-in point B and can be found by numerically solving the following set:

$$\begin{cases} \dfrac{2\kappa\lambda^2(1-\lambda^2)}{1+\kappa\lambda} - 4\lambda + \left[\dfrac{\kappa\lambda(1-\lambda^2)}{1+\kappa\lambda} + 2D - \ln(1+\kappa\lambda)\right]\dfrac{2}{\kappa D}\ln(1+\kappa\lambda) = 0; \\ \beta = \left(2\lambda - 2\dfrac{\ln(1+\kappa\lambda)}{\kappa} + \dfrac{\ln^2(1+\kappa\lambda)}{D\kappa}\right)\dfrac{(1-\lambda^2)}{\lambda^2}. \end{cases} \tag{20}$$

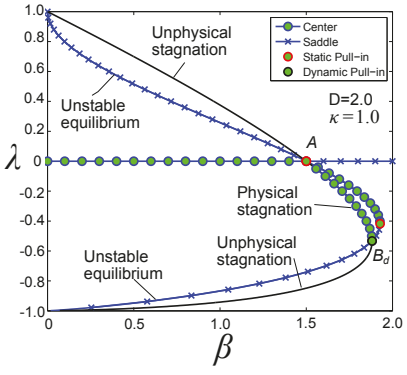


Figure 6. Static and dynamic bifurcation diagrams: solid black lines correspond to unphysical stagnation.

Similar to Section 4, we consider the case when κ is small, then the set (20) has an approximate solution:

$$\lambda_{B_d} \approx -\kappa \frac{D/3+1/2}{D+1}; \;\; \beta_{B_d} \approx \frac{\kappa}{D}\left(1-\kappa^2 \frac{(D/3+1/2)^2}{(D+1)^2}\right). \tag{21}$$

Note that, when the spacing κ tends to zero, the static and dynamic pull-in displacements tend to their zero initial position, and pull-in voltages also tend to zero. Once $\kappa = 0$, all static and dynamic pull-in points merge into one zero point.

6. Conclusions

In this article, a micro-machined hybrid suspension based on combining electromagnetic inductive and electrostatic actuation, which provides, in particular, control over and decrease of the vertical component of stiffness, was presented. We discussed the micromachined fabrication process of the device establishing three micro-structures, namely coil, top electrode and bottom electrode structure. In particular, two possible ways of fabrication of the bottom electrode structures based on SOI and Si wafer were considered. Using the developed micro-machined process, a prototype of the suspension was successfully fabricated. The preliminary experimental study of this prototype was performed and successfully demonstrated the proof of concept of the device proposed in [30]. In particular, the disk-shaped aluminum proof mass was levitated between the top and bottom structures generating the electrostatic field. A decrease of the vertical component of the stiffness by 25% was successfully observed.

A generalized analytical model of the suspension was also developed. In order to study the behavior of the hybrid suspension along the vertical direction as a particular case of the developed model, a quasi-exact nonlinear model was formulated. Then, using the design particularities of a micro-machined version of the suspension, a reduced model was proposed. It is worth noting that the applicability of the analytical technique used for modeling a hybrid suspension has been already successfully demonstrated for instance in [32]. Using the reduced nonlinear model, the static and dynamic responses of the suspension were analytically and comprehensively investigated, and the static and dynamic pull-in parameters were identified. In particular, it was shown that within the framework of the developed prototype, the initial stiffness generated by the inductive suspension could be reduced by two orders of magnitude.

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Author Contributions: Kirill V. Poletkin developed and fabricated the device, performed the experiment, developed and studied the models and wrote the manuscript. Jan G. Korvink analyzed intensively the developed model and the obtained experimental results and also contributed in improvement of the writing of the manuscript.

Conflicts of Interest: The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AM	Amplitude modulator
DRIE	Deep reactive-ion etching
NS	Negative stiffness
SIO	Silicon-on-Insulator
μ-CS	Micro-machined Contactless Suspensions
μ-ECS	Micro-machined Electrostatic Suspensions
μ-MCS	Micro-machined Magnetic Suspensions
μ-HCS	Micro-machined Hybrid Suspensions

Appendix A

The accuracy of approximation of modeling the electromagnetic force by means of the reduced model (8) depending on the parameter $\xi = h_l/(2r_l)$, as shown in Figure A1. Analysis of Figure A1 shows that, if the parameter ξ is less than 0.3, the electromagnetic force is approximated by the logarithmic function (6) with an error less than 6%.

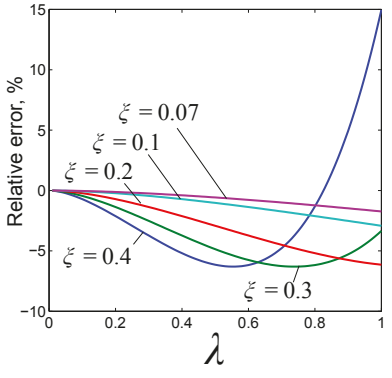


Figure A1. Accuracy of modeling the electromagnetic force by means of the reduced model (8) as compared to the quasi-exact model (5).

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Learning Micromanipulation, Part 1: An Approach Based on Multidimensional Ability Inventories and Text Mining

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Abstract: In the last decades, an effort has been made to improve the efficiency of high-level and academic education players. Nowadays, students' preferences and habits are continuously evolving and so the educational institutions deal with important challenges, such as not losing attractiveness or preventing early abandonment during the programs. In many countries, some important universities are public, and so they receive national grants that are based on a variety of factors, on which the teaching efficiency has a great impact. This contribution presents a method to improve students commitment during traditional lessons and laboratory tests. The idea consists in planning some activities according to the students' learning preferences, which were studied by means of two different approaches. The first one was based on Gardner's multiple intelligence inventory, which is useful to highlight some peculiar characteristics of the students on the specific educational field. In the second method, direct interviews, voice recognition, and text mining were used to extract some interesting characteristics of the group of students who participated in the projects. The methods were applied in May 2018 to the students attending the course of Micro-Nano Sensors and Actuators for the postgraduate academic program dedicated to Industrial Nanotechnologies Engineering of the University of Rome La Sapienza. The present paper represents the first part of the investigation and it is dedicated essentially to the adopted methods. The second part of the work is presented in the companion paper dedicated to the presentation of the practical project that the students completed before the exam.

Keywords: teaching; learning; mocromanipulation; microactuation; multiple ability inventories

1. Introduction

Teaching methods and procedures at the high level of education and in Academia have been changing during the last years, whereas improving the efficiency of the educational process has become a crucial target for institutions [1,2]. Related studies show that improving the teaching–learning process has a certain impact on economic development [3], while the process itself is ineluctably affected by the mutual influences between technology and society [4] and by socio-emotional issues [5].

Furthermore, in many European countries, recent laws apply rewarding criteria to those institutions who support innovation in teaching and reduce the number of students abandoning or prolonging their courses.

In this paper, a recent experience of teaching–learning activities is presented, with the purpose of suggesting a way to improve students involvement and achievements. The adopted methods were applied in May 2018 to the students attending the course of Micro-Nano Sensors and Actuators for the Master degree in Industrial Nanotechnologies Engineering of the University of Rome La Sapienza. The present paper is dedicated essentially to the description of the teaching methods, whereas the consequent students' activities are presented in its companion paper [6].

An Approach Based on Multidimensional Ability Theoretical Framework

The basic idea for this investigation consisted in detecting the students' learning preferences and then tailoring some activities that could match the detected learning styles. This approach was expected to have the maximum effect in terms of achievements and satisfaction. The detection of the students' preference profile was approached by means of two different methods. The first one was inspired by the concept of Multidimensional Abilities, while the second one was based on text analysis.

Multidimensional Abilities have been investigated through well-known theoretical frameworks:

- Aptitude-Treatment Interaction [7];
- Conditions of Learning [8];
- Dual Coding Theory [9];
- Multiple Intelligences [10];
- Structure of Intellect [11]; and
- Triarchic Theory [12].

In this study, Gardner's approach was preferred because it has been applied in the past and gave some interesting results (see, for example, [13–16]). According to the original formulation of Multiple Intelligence inventory, the following intelligences are considered:

- Intrapersonal;
- Bodily-Kinesthetic;
- Interpersonal;
- Logical-Mathematical;
- Spatial;
- Verbal-Linguistic; and
- Musical.

Nowadays, Gardner's Multiple Intelligence is very popular and its principles have been described in many reference books and websites. In the present investigation, this inventory was applied to the specific context of learning in a technological education field, micromanipulation, in a high-level academic course. Consequently, the investigation must be considered as restricted to a rather superficial layer of one student's individual personality. The results do not concern the individual's general characteristics, but rather involve the individual's characteristics in the context of the specific environment, namely, a course in Industrial Nanotechnologies Engineering. To better explain this concept, let us review how the multiple intelligences have been applied in the context of learning in the academic environment.

Intrapersonal intelligence was not measured as an individual's general characteristic, but as a student's attitude of thinking and studying alone, without external interactions.

Bodily-Kinesthetic intelligence was associated with the use of practical activities that involve manual handling of models or instruments in a lab. This attitude was also associated with the use of physical support for studying and understanding.

Interpersonal intelligence was here related to some students attitude to gain benefit from social activities, such as studying in the company of other people and participating or promoting small social events.

Logical-Mathematical intelligence was associated with the students' preference for using logic and mathematical formulation to understand new models and concepts.

Spatial intelligence was related to a student preference for using graphical tools, such as pictures, diagrams, block charts, or spatial models.

Verbal-Linguistic intelligence was related to the students' preference of learning by reading or hearing some text.

Finally, Musical intelligence was measured as in standard tests but was not related to any dimension for the sake of the present investigation.

2. Material and Methods

A questionnaire was submitted to a limited group of students to measure their singular preference profiles based on Gardner's inventory. The adopted test has the advantage that it has been used before [16] on a reference group and on some groups of students enrolled in the courses of Engineering at Sapienza University. Therefore, averages and standard variations on raw scores were already available for the adopted dimensions. This circumstance was quite important because the number of participating students was rather limited ($n = 6$).

2.1. Correlation Analysis among Dimensions

As for the previous investigations [14–16], standard statistical analysis was applied to the raw scores obtained from the surveys. Then, standard z-scores were calculated by an elementary routine. z-scores are a standardization of the raw data which maps scores in such a way that the standardized population has a null mean and a unit standard deviation (see Figure 1).

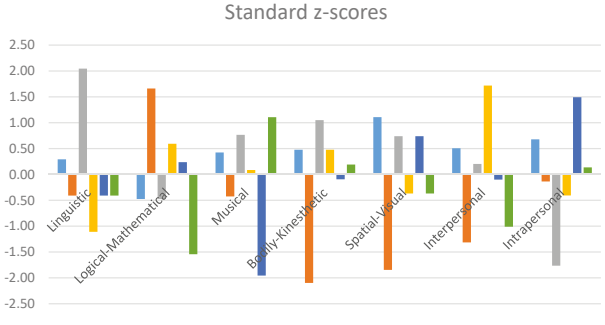


Figure 1. Individuals standard z-scores for the adopted seven dimensions.

The adoption of the z-scores was useful because the calculation of the correlation matrix becomes quite easy. Table 1 shows the classical correlation table for the seven dimensions under analysis. Since only six individuals participated in the survey, the obtained results are not significant for possible generalization and so the curious positive or negative correlations have a value restricted to the analyzed group.

Table 1. Correlation table for the group of participating students ($n = 6$).

	Verbal	Logical	Musical	Kinesthetic	Spatial	Interpersonal	Intrapersonal
Verbal	1.00	−0.32	0.35	0.43	0.47	−0.06	−0.59
Logical		1.00	−0.57	−0.71	−0.57	0.01	0.03
Musical			1.00	0.39	−0.01	0.03	−0.63
Kinesthetic				1.00	0.81	0.64	−0.26
Spatial					1.00	0.47	0.15
Interpersonal						1.00	−0.14
Intrapersonal							1.00

Figure 2 shows the scatter diagrams which illustrate an example of both positive and negative correlations among two pairs of dimensions. As mentioned above, it is not possible to draw a general conclusion that any student who has a kinestetic preference must have necessarily a great spatial-visual attitude and a small Logical-Mathematical preference. This is only a trend that appeared in a small class.

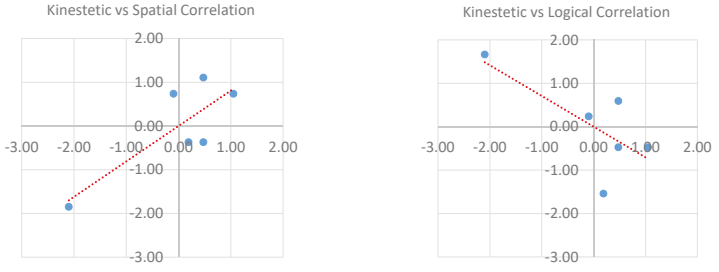


Figure 2. Scatter diagram for two pairs of dimensions: (**a**) Spatial-Visual vs. Bodily-Kinesthetic positive correlation; and (**b**) Spatial-Visual vs. Logical-Mathematical negative correlation.

2.2. Characteristics of the Class Group

The present investigation attempted to ascertain whether the small class was characterized by peculiar characteristics. Of course, this analysis must be done by comparing the dimensions averages of the class with those obtained in a control group. Fortunately, the same questionnaire had been previously submitted to a much larger control group during an earlier investigation [16], and so these results were used for the present analysis. For each of the seven dimensions, the pairs of averages were compared by pointing out those with a minimum value for the significance levels. Hence, the null hypothesis $\mathcal{H}_0$ was set as the assumption that the class and control groups were extracted from the same population.

Weak ($0.01 < \alpha \leq 0.1$) or strong ($0.001 \leq \alpha \leq 0.01$) probability levels were adopted together with Student t-test and the degrees of freedom $n_A = 6$ and $n_B = 56$ corresponding to class and control group populations, respectively. Some significant differences are reported in Table 2.

Table 2. Significant differences between raw score dimensions of the class and control group populations.

	Verbal	Logical	Musical	Kinestetic	Spatial	Interpersonal	Intrapersonal
	lower	greater	lower	greater	greater	greater	lower
p	>0.1	0.025	>0.1	0.1	0.001	0.001	>0.1
	no	weak	no	weak	strong	strong	no

- Verbal (or Linguistic) Intelligence: According to previous investigations, students enrolled in high-level Engineering courses present verbal capabilities usually higher than their peers who

do not frequent Academic Institutions. Our class students present the same characteristic as the control group students, and therefore no significant difference was found for this dimension.

- Logical-Mathematical Intelligence: Our students showed a preference for logical and mathematical formalism weakly greater than the students belonging to the control group. This is quite reasonable for students enrolled in the last year of a master course in Nanotechnology.
- Musical Intelligence: No significant difference was detected for this dimension.
- Kinesthetic Intelligence: This dimension is characterized by a weakly significant difference in the average scores for the two groups; actually, engineering students are quite handy and so our students simply showed an attitude a little more pronounced than the control group.
- Spatial-Visual Intelligence: According to the significance analysis, it is quite sure that our students prefer an approach based on practical observation of planar schemes and 3D objects manipulation more than peers students enrolled in other courses; this dimension suggests the use of real objects and the practical manipulation of real models as an effective teaching method for this class.
- Interpersonal Intelligence: The clear preference in the class of an approach based on interpersonal intelligence is a straightforward suggestion to adopt interactive lessons with a considerable amount of teamwork.
- Intrapersonal Intelligence: No significant difference with respect to the control group was detected for this dimension.

3. Text Mining for the Identification of Patterns

Before the starting of the course, some interviews were arranged with the students. The transcripts of the oral interviews were automatically obtained by means of available software with the simple aid of a smartphone.

The following questions were submitted to the students before the beginning of the course.

1. Which is your personal motivation for you to study Nanotechnology?
2. What do you think about the activities in the lab?
3. What is your opinion about manual activities?
4. Do you think they can help your understanding of the lessons?

The following questions were submitted to the same students during of the course.

5. Did you change your opinion about the activities in the lab?
6. How much do you feel adequate for this experience in the lab?
7. What are your personal features that you think they are the most suitable for this experience?

These questions were used as cognitive stimulus to trigger the explication of concepts from the interviewees. The questions had the function of making some concepts more visible than in the case of a free speech. The advantage of this technique consists in the possibility of handling quantitatively some parameters of interest.

Then, the answers were filtered to remove from text some insignificant part of the spoken language and to obtain a polished text (conjunctions, prepositions, articles, and so on). The latter was then analyzed using text mining classification and frequency analysis of words occurrences. The graphic technique of words cloud was applied to extract key concepts and the most used words, and a graphic composition of such collection is reported in Figure 3, where dimensions and colors are related to the occurrence frequency. This wordcloud representation is very interesting to gain a picture of the tags that were most raised during the conversations with students.

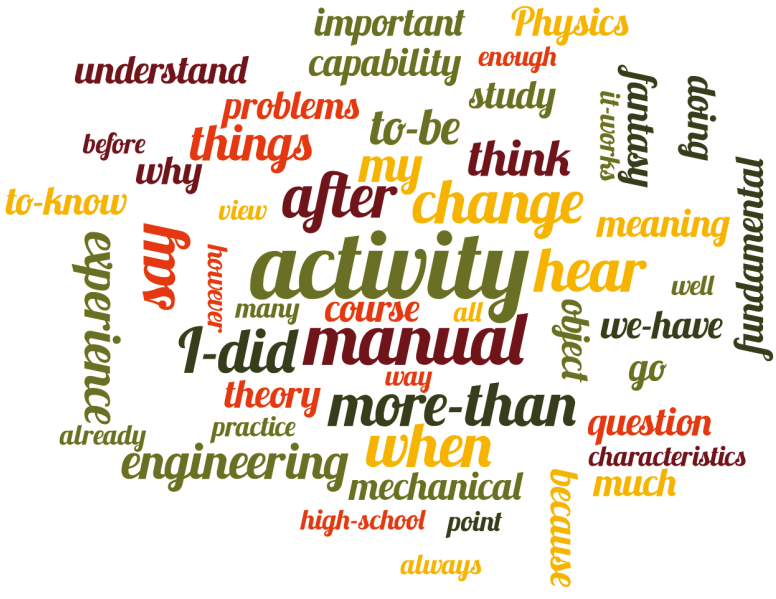


Figure 3. Most frequently mentioned words during interviews.

The following are some examples of frequently spoken words during the interview (see also Figure 3), and some reasonable interpretations.

- activity, experience, things, practice, hand, it-works, manual, and doing: These words are quite central during the interview; they are related to the students direct involvement to the lab activities and their willingness to learn by means of direct experience rather than by means of the classical textbook study.
- view and hear are also related to senses and so to the experience.
- more than, when, after, before, change, and go: These words are clearly related to time, development and progress and therefore they show the students willingness to improve, compete and face challenges.
- engineering, mechanical, Physics, high school, and study: These are clearly related to the context of the course.
- think, understand, meaning, fundamental, and important: These are related to cognitive and learning process.
- my and I-did: These are probably related to the students positive self promotion, and to the concept of making valuable their own experience.
- fantasy, finally and curiously, appears as a possible sign of passion for the topic and a windows on creativity in design.

The same text mining technique was applied to the course syllabus and the most cited words are presented in a graphical word cloud (Figure 4). A comparison of the two clouds suggests that the main differences consist of terms that represent action and personal experience. Once again, the comparison shows how this class students of were sensible to the manual and practical activities while learning a topic which implies, intrinsically, some theoretical concepts. In fact, micro and nano actuation is a topic which requires multiple (and rather opposite) abilities such as theoretical and practical skills. This is a possible reason for the success of the practical projects reported in the second part of this investigation [6].

The graphics in Figures 3 and 4 were generated by means of Wordsalad © free app [17].

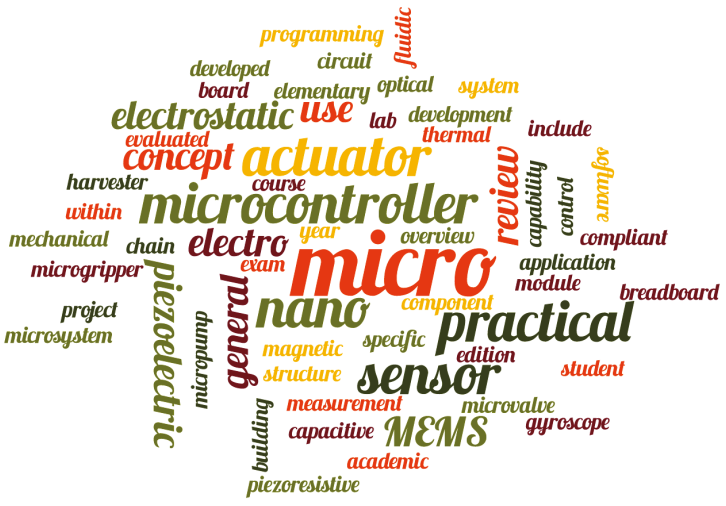


Figure 4. Most frequently mentioned words in the course syllabus.

4. Results and Discussion

Considering the significance analysis described in the previous sections, our class presents three significant differences with respect to the control group: Spatial-Visual, Interpersonal and Kinesthetic Intelligences. The results obtained from the text analysis also showed an evident inclination of the students to the activities and practical work and therefore they confirmed the results obtained by using Gardner's inventory. Therefore, considering that the Course basic material is based on classical text books [18,19] and on some recent papers [20–31], i.e., written material, the peculiar above-mentioned results gave rise to the idea of encouraging our students to work in team (Interpersonal) and make use of handmade (Kinesthetic) models (Spatial-Visual).

Since this is not possible for nano or micro size devices, the idea of using properly up-scaled models arose. By assembling the indications from the Multiple Intelligences method and from the text analysis together, it was clear that some practical activities would have given a great boost to teaching and therefore some projects were designed for the specific group of students.

These projects have the following characteristics.

- From the course program, a micro or nano device was selected as the case study.
- A series of activities is planned for the selected device, according to the common operational capability required for the real micro or nano system under study.
- A macroscale model is obtained for the micro or nano device by using 3D printing low-cost techniques.
- The operational features are translated from the micro- (or nano-) to the macroscale, possibly by changing some principles of operation (this allows the students to understand some fundamental concept of scaling in fabrication and actuation).
- During these activities tinkering is encouraged for the construction of the system at the macroscale.
- Finally, a system is created at the macroscale based on the micro- or nanosystem, and the differences between the two systems are material for the exam.

Some of these projects are described in detail in the companion paper [6].

Considering the excellent scores achieved by the students during the exams and their great satisfaction, it is possible to conclude that the presented teaching–learning experience had quite a positive impact on the course. More generally, results show that, thanks to the adopted methods, the quality of teaching/learning can be improved also in the context of Nanotechnology and Micro Actuation.

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Learning Micromanipulation, Part 2: Term Projects in Practice

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Abstract: This paper describes the activities that have been necessary to design, fabricate, control and test some low-cost test stands independently developed by the students enrolled in the course of Micro-Nano sensors and actuators for the postgraduate course in Industrial Nanotechnologies Engineering of the University of Rome La Sapienza. The construction and use of these test stands are an essential part of teaching and learning methods whose theoretical bases have been presented in the companion paper (Part 1). Each test stand is composed of a compliant structure and a control system, which consists of a programmable control micro-card equipped with sensors and actuators. The compliant structure consists of a compliant mechanism whose geometry is achieved by scaling some previously developed silicon micromanipuators and microactuators up to the macroscale by a factor of 20. This macroscale model offered a kinesthetic tool to improve the understanding of the original microsystems and their working principles. The original silicon micromechanisms have been previously presented in the literature by the research group after design and deep reactive-ion etching (DRIE) microfabrication. Scaling from micro to macro size was quite easy because the original DRIE masks were bestowed to the students in the form of CAD files. The samples at the macroscale have been fabricated by means of recently available low-cost 3D printers after some necessary modifications of the mask geometry. The purpose of the whole work (Parts 1 and 2) was the improvement of the efficiency of an educational process in the field of microsystem science. By combining the two companion papers, concerning, respectively, the theoretical basis of the teaching methods and the students' achievements, it is possible to conclude that, in a given class, there may be some preferred activities that are more efficient than others in terms of advancements and satisfaction.

Keywords: teaching; learning; mocromanipulation; microactuation; multiple ability inventories

1. Introduction

This paper presents some activities that have been carried out during the class of Micro-Nano sensors and actuators for the postgraduate academic program dedicated to Industrial Nanotechnologies Engineering of the University of Rome La Sapienza. The first part of the investigation [1] presented two didactical methods, the first one being based on Gardner's multiple intelligence inventory and the second one based on text analysis. Multiple intelligences have been used to reveal a characteristic profile of each student and some significant characteristics of the whole group. Secondly, text analysis has been applied to the transcript of some interviews with the students and also to the course syllabus, as a reference text for comparison. The two methods showed that this particular class had a strong inclination to the practical and interpersonal work. Based on those results, the teaching staff decided to stress the importance of term projects because they allow students to get easily familiar with the micro world through their preferred kinesthetic approach and motivational feelings. The present paper describes the projects that have been carried out by the students their self during

the semester. This preferences–based teaching–learning method has the advantage of increasing the course attractiveness, participation and efficiency in terms of students achievements.

Since the students participate in the above-mentioned course during their last year of the Master degree, the teaching material must be quite upgraded and also at the scientific literature level, because micro and nano technology is rather rapidly changing. Other than studying some portions of classic text books on micro and nano technology [2,3], more specific readings were suggested, related to the previous experience achieved by the teaching staff and concerning compliant 3 D.O.F. microrobots [4–7], micromechanisms [8], micro hinges [9–12] and microgrippers [13–18]. However, this material is essentially written text.

As explained in the companion paper [1], the verbal approach is not the only way students learn. For the class group involved in the present investigation, some kinesthetic activities have been encouraged during laboratory meeting. The idea of increasing students–object tactile interactions comes from the fact that course subject consists of micro and nano objects, which are even hardly visible. Therefore, there are few ways of presenting them: showing the masks at a certain magnification, producing SEM or optical images or short animations of the objects, or through verbal description. In all these cases, there is scarce direct or manual interaction. Nevertheless, it has been shown that kinesthetic exercises may activate human original corporeal–kinetic apprenticeship [19].

During lab activities, the students have been provided with a certain number of silicon prototypes for observation and analysis. However, each sample consists in a device that is embedded within a 2 square millimeter, overall size, with some specific parts, such as the comb drive fingers, which have two-micron width. This makes direct interaction practically impossible. Bringing the microsystem to the macroscale is useful not only for simulation but also for teaching purposes. In fact, the students become able to develop their own simplified systems that can be also operated at the macroscale by means of inexpensive micro-cards. These activities can be also useful to stimulate a certain capability of working in team. Finally, thanks to the development of new products and to some new methods based on tinkering [20], the overall cost of these lab activities can be extremely limited, if compared to the usual costs of equipment used in the common nanotechnology lab, while the students involvement remains very high. The term projects described in the next section have been developed by the students under the surveillance and advice of the teaching staff.

2. Material and Methods

2.1. Fabrication of the Mechanical Structure

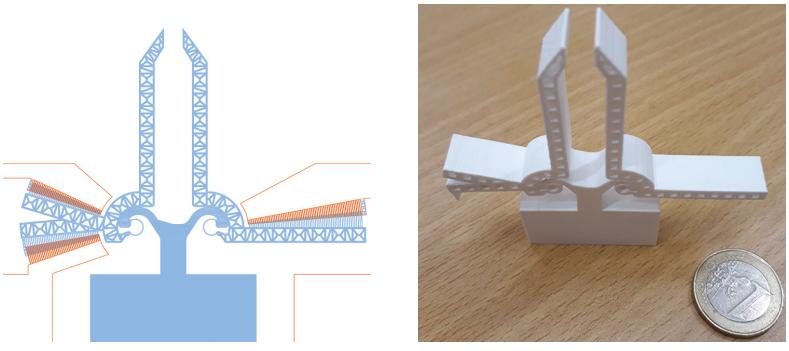
The widespread success of additive manufacturing made 3D printers available at low costs. Many students and practitioners have been able to assemble their own 3D printers by using limited resources and this circumstance made these devices very popular among students. For this reason, the idea of using a 3D printer for scaling up a microsystem, although not very original, has been accepted with a certain enthusiasm by the students.

The students were first provided with the original mask CAD files that have been used to fabricate the real microsystems. Then, they modified the 2D geometry to adjust the beams or the fingers minimum width to the maximum resolution of the adopted printer. The 3D printer was also used to fabricate the housing for the compliant mechanism, the motors, the micro-cards and all the necessary mechanical parts, in such a way that the whole system would be correctly assembled on the frame.

2.2. Double Rotary Link Microgripper

The first project consisted in the fabrication and the experimental operation at the macroscale of a simple double rotary link gripper. The students were provided with the original mask CAD files for the DRIE (Deep Reactive Ion Etching) process (see Ref. [14] for details). The actual design was developed by means of AutoCAD© (Version 2015, Autodesk, San Rafael, CA, USA), and so the starting point, for the students, was the mask depicted in Figure 1a. This design was scaled as well as

modified to fit the limits of the 3D printer accuracy to avoid unfeasible widths of fingers, gaps and curved beams. The truss structure embedded in the original links was also simplified, in order not to make critical the process of 3D printing. Finally, the fingers of the mobile comb drives were eliminated for the sake of simplicity.



(**a**) (**b**)

Figure 1. The original mask for a compliant rotary microgripper (**a**); and its prototype at the macroscale (**b**) (first project).

The AutoCAD *dwg* file was translated into the classical *stl* file and then provided to the 3D printer, which yielded the layout represented in Figure 1b.

The macroscale compliant mechanisms were subject to preliminary tests which gave encouraging results in terms of mechanical properties (compliance and robustness). Then, the experimental layout was set for operational tests. Since the comb drives were not feasible at the macroscale, their action was simulated by means of cables attached to the end of the comb drive mobile wings and operated by a motor. A simplified scheme is reported in Figure 2, where the two wings, the two cables and the two motor arms are represented, respectively, by lines AB and DE, BC and EF, and CO and FO. When the cables are under tension, the groups ABCO and DEFO can be considered as four-bar linkages, which makes it very easy to perform the kinematic analysis of the system. In the case under study, the forward run was achieved by applying a counterclockwise torque, while the return run was obtained simply by releasing the above mentioned torque, by virtue of the elastic energy stored in the flexure revolute joints A and D.

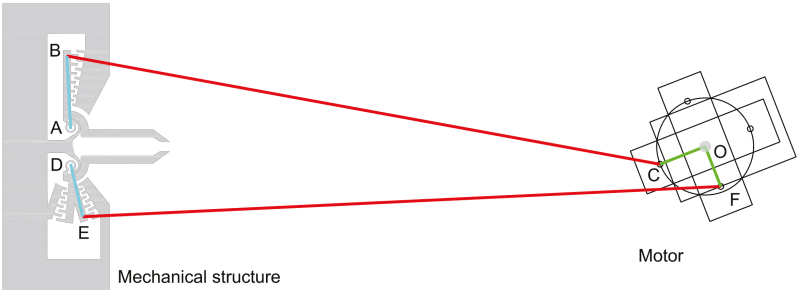


Figure 2. Actuation of the compliant mechanisms at the macroscale (first project).

The experimental layout was completed using the same 3D printer for building the mechanical anchors and frames. The printing parameters were set by using Ultimaker Cura software©

(Version 3.2.1), and the G-code file was prepared. All components were re-printed several times and then assembled to the base. Shaking torques transmitted to the base were estimated and a proper frame was set on the motors to allow them to work adequately. Electronic circuits and control card were mounted to the same base, as a reminder that the real microsystem would be arranged on a single chip, by using ASIC (Application Specific Integrated Circuit). The arrangement is illustrated in Figure 3.

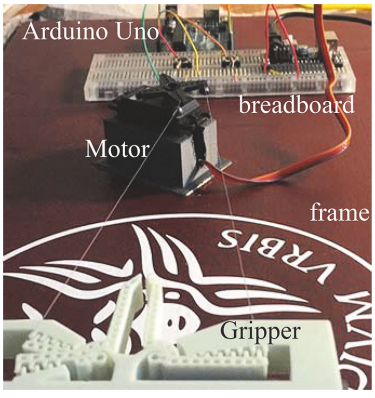
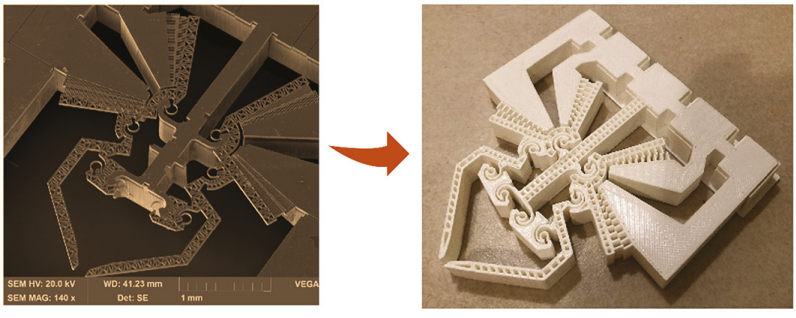


Figure 3. A picture of the experimental layout (first project).

The actual circuit was developed using Fritzing© (Version 0.9.3), and the wiring scheme was realized by using a common breadboard, which includes also connection to Arduino UNO© (REV3). As is usual for this particular microcontroller, a simple Arduino IDE© code (Version 1.8.5), was developed, tested and uploaded on the microcard, in such a way that the whole system no longer needed direct connection to the computer. Forces were transmitted to the compliant mechanisms through nylon cables. In the first configuration, the closing motion was obtained by exerting a force on the jaws levers, while the opening return motion is achieved by releasing these forces and letting the flexure elasticity to drive back the jaws to their original position.

2.3. Bidirectional Double Four-Bar Linkage Microgripper

In the second project, a bidirectional microgripper, shown in Figure 4a, was scaled up to the macroscale sample, represented in Figure 4b, by following the same method of design, printing and actuation as described in the previous paragraph. The finger cross section was scaled from 2 μm × 40 μm to 0.6 mm × 1 cm, while the microgripper overall dimensions are 10.8 cm × 10.3 cm × 1 cm. The macroscale prototype was obtained in Polylactic Acid, while actuation and control was obtained by Arduino microcontroller integrated with breadboard, circuits and two rotary actuators. The comb drives were mechanically controlled by means of cables that are attached in such a way that open and close maneuvers are allowed.



(**a**) (**b**)

Figure 4. Comparison of the micro (see Ref. [14]) (**a**) and macro (**b**) grippers (second project).

The main problem with design consisted of the limited resolution of the 3D Delta printer. With the available model, the resolution was limited to 0.4 mm, thus some sections of the gripper had to be redesigned to cope with this limit. In fact, after linear scaling from the micro to the macro scale, there were still some parts (fingers and curved beams) that were too small for that resolution. Furthermore, it was also not possible to use the above-mentioned minimum size to not slow down the printing process too much, and so the minimum thickness had to be increased to 0.6 mm for both finger and gap between them. AutoCAD 2018 student version was used for mask scaling and redesigning (estimated time, non-expert user, 30 h for one person).

Ultimaker Cura© (version 3.2.1), was used for managing the printing process and changing some printer properties and maximize the likelihood of success. The main adopted changes in the printer properties were: layer height (0.1 mm), wall thickness (0.4 mm), printing temperature of the nozzle (220 °C), build plate temperature (30 °C) and print speed (35 mm/s), while cooling was enabled and the build plate adhesion type set to Skirt. After the printing stage, some surface grinding operations were performed to polish the surfaces (estimated time, non-expert users, 20 h for one person).

Similar to the first project, the system at the macroscale was made independent of the computer by using Arduino UNO© capability of uploading IDE codes ready to work with the system. The actuation scheme is represented in Figure 5. In this case, there are two motors which pull the comb drive wings to open the jaws. With reference to Figure 5, cable FE pulls body CDE to rotate around flexure center D, in such a way that the coupler CB drives the attached jaw toward the closure position.

The motors were controlled by means of a circuit that is similar to that described in the previous paragraph, being composed of a breadboard, some simple electronic components and a microcontroller. A picture of the experimental set is reported in Figure 6.

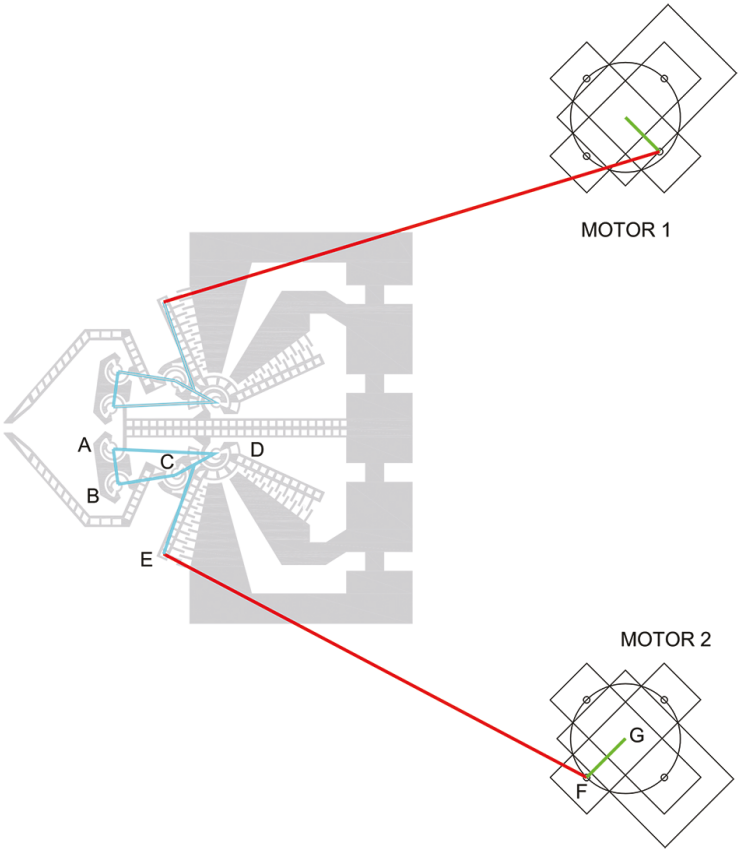


Figure 5. Actuation of the compliant mechanisms at the macroscale (second project).

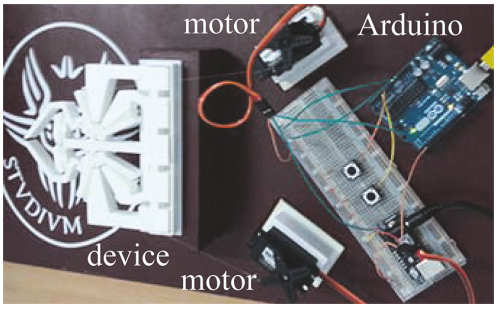


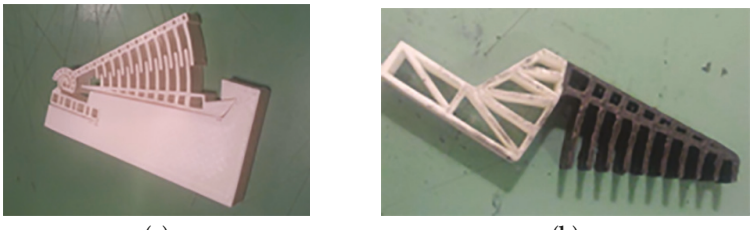
Figure 6. A picture of the experimental layout (second project).

3. Results and Discussion

The projects described in the previous section are useful mainly for educational purposes. In fact, the final layout is very similar to a chip that could be used for the real microsystems. The experimental scaling was also important for the student to understand that actuation is strongly related to size. The macroscale grippers were actuated using classical electric motors, while the microgrippers were

actuated by means of comb drives. Nevertheless, the projects, as a whole, were also interesting for research purposes, because they were useful to point out all the necessary operations and components.

The attempt to build a system uniformly scaled from micro- to macroscale was used to appreciate some basic and very well known concepts of actuation at the microscale. In fact, the students were encouraged to build a comb drive at the macroscale similar to the one depicted in Figure 7a,b, which they did, of course, with no success. Therefore, the real actuation at the microscale was attempted by arming the wings of two symmetric magnetic circuits (see Figure 8a) with conductive media and then providing magnetic fields to the base of the system to achieve motion (see Figure 8b).



(**a**) (**b**)

Figure 7. Unsuccessful comb drive at the macroscale: (**a**) bulk; and (**b**) covered by conductive paste.



(**a**) (**b**)

Figure 8. Magnetic actuation at the macroscale: (**a**) bulk *stl* file (left and right wings); and (**b**) motion sequence of the armed wings.

These two simple experiments stimulated the discussion concerning the effects of scaling on actuation. In particular, the students were encouraged to figure out the consequences of a uniform reduction of the geometry by a factor of 10. As known, such reduction gives rise to a reduction of the electrostatic force by 100 times, while the corresponding reduction of the curved beam rotary stiffness is 1000. Considering that the electromagnetic force is reduced 10,000 times, it is clear how electrostatic actuators perform much better than the electromagnetic at the microscale.

The benefit of these activities was not limited to the specific course, but it was also extended to more general skills, such as working in a team, rapid prototyping, actuation, control and mechanical design. In particular, the students had to solve specific problems that are of a certain interest today.

The first problem was the optimal setting of the 3D printing system parameters, by taking into account some specific properties such as the nozzle section (0.4 mm) and the material (Polylactic acid, PLA). Then, the operational parameters had to be considered, such as the printing temperature (210–215 °C), and speed. Furthermore, different actuation strategies were analyzed and compared. After exclusion of comb drives, for losing efficacy after dimensional scaling, the servomotors power supply was originally arranged directly from one Arduino 5 V pin. This solution gave rise to several problems, such as torque ripples, unstable position and overheating due to excessive current. Therefore, the classical external power supply scheme with grounding GND pin has been arranged for both the motors, while Arduino provided control. Finally, safety principles have been also implemented during the activities, especially by means of personal protective equipment (PPE).

4. Conclusions

This paper has shown a way to improve the teaching activities in the field of microsystems and micro-manipulation. The investigation has proven how the complex teaching–learning process may be improved by tuning lab activities to the specific students' characteristics, expressed in terms of multiple intelligence inventory, interest, and motivation. The investigation has also proven that the use of low-cost, tinkering approach boosts the student involvement up to a level of interdisciplinary knowledge which was good enough for them to build a system complete with mechanical structure, transmission, actuation, and control. Students declared their satisfaction and excitement during the project, while the final exams showed that their understanding of phenomena at the microscale improved much after the tactile activities at the macroscale. The authors hope that the two papers will be somehow useful to professors for employing new methods of interactive and dedicated teaching for their laboratory courses.

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